

92. Let the ten's digit be x and unit's digit be y .

$$\text{Then, } (10x + y) - (10y + x) = 36$$

$$\Leftrightarrow 9(x - y) = 36 \Leftrightarrow x - y = 4.$$

93. Let the ten's digit be x and unit's digit be y .

$$\text{Then, } (10x + y) - (10y + x) = 63$$

$$\Leftrightarrow 9(x - y) = 63 \Leftrightarrow x - y = 7.$$

Thus, none of the numbers can be determined.

94. Let the ten's digit be x and unit's digit be y .

$$\text{Then, } x + y = \frac{1}{5}[(10x + y) - (10y + x)]$$

$$\Leftrightarrow 5x + 5y = 9x - 9y \Leftrightarrow 4x = 14y.$$

Thus, the value of $(x - y)$ cannot be determined from the given data.

95. Let ten's digit = x . Then, unit's digit = $(12 - x)$.

$$\therefore [10x + (12 - x)] - [10(12 - x) + x]$$

$$= 54 \Leftrightarrow 18x - 108 = 54 \Leftrightarrow 18x = 162 \Leftrightarrow x = 9.$$

So, ten's digit = 9 and unit's digit = 3. Hence, original number = 93.

96. Since the number is greater than the number obtained on reversing the digits, so the ten's digit is greater than the unit's digit.

Let the ten's and unit's digits be $2x$ and x respectively.

$$\text{Then, } (10 \times 2x + x) - (10x + 2x) = 36 \Leftrightarrow 9x = 36 \Leftrightarrow x = 4.$$

$$\therefore \text{ Required difference} = (2x + x) - (2x - x) = 2x = 8.$$

97. Let ten's digit = x . Then, unit's digit = x^2 . Then, number = $10x + x^2$.

Clearly, since $x^2 > x$, so the number formed by interchanging the digits is greater than the original number.

$$\therefore (10x^2 + x) - (10x + x^2)$$

$$= 54 \Leftrightarrow 9x^2 - 9x = 54 \Leftrightarrow x^2 - x$$

$$= 6 \Leftrightarrow x^2 - x - 6 = 0$$

$$\Leftrightarrow x^2 - 3x + 2x - 6 = 0$$

$$\Leftrightarrow x(x - 3) + 2(x - 3) = 0$$

$$\Leftrightarrow (x - 3)(x + 2) = 0$$

$$\Leftrightarrow x = 3.$$

So, ten's digit = 3, unit's digit = $3^2 = 9$.

$\therefore$ Original number = 39.

Required result = 40% of 39 = 15.6.

98. Let the middle digit be x .

$$\text{Then, } 2x = 10 \text{ or } x = 5.$$

So, the number is either 253 or 352.

Since the number increases on reversing the digits, so the hundred's digit is smaller than the unit's digit. Hence, required number = 253.

99. Since the number reduces on reversing the digits, so ten's digit is greater than the unit's digit.

Let the unit's digit be x .

Then, ten's digit = $(x + 1)$.

$$\therefore 10x + (x + 1) = \frac{5}{6}[10(x + 1) + x] \Leftrightarrow 66x + 6$$

$$= 55x + 50 \Leftrightarrow 11x = 44 \Leftrightarrow x = 4.$$

Hence, required number = 54.

100. Let the two-digit number be $10x + y$.

Then, number formed by reversing the digits = $10y + x$.

Difference of squares of the numbers

$$= (10x + y)^2 - (10y + x)^2$$

$$= (100x^2 + y^2 + 20xy) - (100y^2 + x^2 + 20xy)$$

$$= 99(x^2 - y^2), \text{ which is divisible by both 9 and 11}$$

101. Let the unit's digit be x . Then, ten's digit = $(x - 2)$.

$$\therefore 3[10(x - 2) + x] + \frac{6}{7}[10x + (x - 2)] = 108$$

$$\Leftrightarrow 231x - 420 + 66x - 12 = 756$$

$$\Leftrightarrow 297x = 1188$$

$$\Leftrightarrow x = 4.$$

Hence, sum of the digits = $x + (x - 2) = 2x - 2 = 6$.

102. Let the ten's digit be x and unit's digit be y .

$$\text{Then, } \frac{10x + y}{2} = 10y + (x + 1)$$

$$\Leftrightarrow 10x + y = 20y + 2x + 2$$

$$\Leftrightarrow 8x - 19y = 2 \quad \dots(i)$$

$$\text{and } x + y = 7 \quad \dots(ii)$$

Solving (i) and (ii), we get : $x = 5, y = 2$. Hence, required number = 52.

103. Let the ten's digit be x .

Then, unit's digit = $2x + 1$.

$$[10x + (2x + 1)] - [10(2x + 1) + x] - \{10x + (2x + 1)\} = 1$$

$$\Leftrightarrow (12x + 1) - (9x + 9) = 1 \Leftrightarrow 3x = 9 \Leftrightarrow x = 3.$$

So, ten's digit = 3 and unit's digit = 7. Hence, original number = 37.

104. Let the ten's digit be x and unit's digit be y .

$$\text{Then, } 10x + y = 3(x + y) \Rightarrow 7x - 2y = 0 \quad \dots(i)$$

$$10x + y + 45 = 10y + x \Rightarrow y - x = 5 \quad \dots(ii)$$

Solving (i) and (ii), we get : $x = 2$ and $y = 7$.

$$\therefore \text{ Required number} = 27.$$

105. Let the ten's and unit's digits be x and $\frac{8}{x}$ respectively.

$$\text{Then, } \left(10x + \frac{8}{x}\right) + 18 = 10 \times \frac{8}{x} + x$$

$$\Leftrightarrow 10x^2 + 8 + 18x = 80 + x^2$$

$$\Leftrightarrow 9x^2 + 18x - 72 = 0$$

$$\Leftrightarrow x^2 + 2x - 8 = 0$$

$$\Leftrightarrow (x + 4)(x - 2) = 0$$

$$\Leftrightarrow x = 2.$$

So, ten's digit = 2 and unit's digit = 4. Hence, required number = 24.

106. Clearly, there are 4 such numbers: 161, 242, 323 and 404.

107. Let hundred's digit = x .

Then, ten's digit = $(x + 1)$.

$$\text{Unit's digit} = 75\% \text{ of } (x + 1) = \frac{3}{4}(x + 1).$$

$$\therefore (x + 1) + x = 15 \Leftrightarrow 2x = 14 \Leftrightarrow x = 7.$$

So, hundreds' digit = 7; ten's digit = 8; unit's digit = $\frac{3}{4}$

$$(7 + 1) = \frac{3}{4} \times 8 = 6.$$

Hence, required number = 786.